

Todd Takala

Lead Data Scientist — Lead Machine Learning Engineer — AI/ML Engineer — Remote

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Professional Summary

Motivated Lead Data Scientist with a proven track record of delivering over \$1.3 billion in business value through machine learning and end-to-end data science solutions for complex challenges. Expert in managing the software development lifecycle, emphasizing stakeholder engagement, KPI definition, team building, and scalable deployment. Ability to articulate complex data insights to diverse stakeholders, ensuring clear understanding and informed decision-making.

Experience

Lead Data Scientist — Caterpillar Inc. — Surprise, AZ — 09/2022 to Present

- Architect and developer of end-to-end high-impact data science initiatives, including an award-winning AI platform to predict equipment repair times, resulting in an \$80 million reduction in warranty costs and a CEO Award.
- Possessing business acumen to collaborate with senior leadership and stakeholders across the organization which facilitated an understanding of the project's broad impact, and the creation of precise user requirements. This ensured that the project remained aligned with the business road map through both written and persuasive verbal communication.
- Wrangled structured, semi-structured and unstructured data to assemble a training dataset from multiple sources. Thousands of parameters were considered derived from dealer and product information. An LLM was used to find similarities in job descriptions and feature engineering was used to keep important parameters.
- The dataset designated for training was subjected to exploratory data analysis, where outliers were eliminated employing sophisticated analytics and anomaly detection techniques. Python, along with Autogluon, was utilized to train the AI model, capitalizing on machine learning frameworks such as neural networks, PyTorch, XGBoost, TensorFlow, and LightGBM.
- Engineered and deployed robust MLOps pipelines using AWS and Snowflake, as a database administrator, and established best practices for cloud and system architecture. Managed project lifecycles and created comprehensive monitoring plans to validate business impact.

Reliability Analytics Manager — Amazon.com — Tempe, AZ — 02/2022 to 09/2022

- Pioneered design engineering KPI for autonomous robots by analyzing petabyte-scale IoT big data. Authored an influential white paper, as reproducible research using R, on the framework for performing statistical modeling and secured adoption by senior leadership.
- Designed and optimized winning regression and recommendation systems, achieving two-time champion status in Amazon Machine Learning University competitions.

Data Engineering Manager — Empire-Cat — Mesa, AZ — 01/2012 to 02/2022

- Created ML model for engine reliability, durability, repair, preventive maintenance and intervention leading to \$62 million in savings within 90 days.
- Developed and deployed an MLOps pipeline with ARIMA and SVM for AI-driven forecasting in supply chain management, integrated data, managed a component exchange program, collaborated with clients on durability forecasting.
- Directed a high-performance team of up to 8 engineers and functioned as client facing subject-matter-expert for mining equipment. Leadership success by fostering a data-driven approach to solve equipment performance issues as Reliability Engineering Manager before being promoted.
- By applying the DMAIC Six Sigma methodology with complex problem-solving using R and SQL, opportunities were identified to improve operation, manufacturing, and maintenance of a large off-highway truck fleet. This approach yielded a savings of \$1.2 billion for our client over a span of six years. Which dramatically improved customer experience.

Education

Bachelor of Science in Mechanical Engineering – Arizona State University — Tempe, AZ — 06/2006

Certifications

- **Arizona State Board of Technical Registration:** EIT, Mechanical Engineering, Registration No. 10670
- **BerkeleyX:** The Science of Happiness at Work Professional Certificate
- **Canonical:** Ubuntu Linux Professional Certificate
- **CSCMP:** Supply Chain Foundations Demand Planning Professional Certificate
- **Data Camp:** AI Fundamentals Certificate, Certified SQL Associate, Data Scientist Professional Certificate
- **Empire-Cat:** Business Tools for Service Managers, Coaching for Peak Performance, Communicating for Leadership Success, Delegating with Purpose, Lean 6 Sigma Green Belt, Resolving Workplace Conflict, Setting Performance Expectations
- **GitHub:** Career Essentials in GitHub Professional Certificate
- **IUOE Local 150:** Diesel Mechanic, Bulldozers, Class B CDL, Cranes, Forklifts, Heavy Equipment Operator, MSHA, OSHA, Scrapers, Skid-steers, Telehandlers, Wheel Loaders Shielded Metal Arc Welding (SMAW)
- **Komatsu North America:** Certified Technical Communicator
- **Society of Automotive Engineers:** Cost and Finance for Engineers
- **Udemy:** Streamlit for Snowflake Masterclass

Hard Skills

A/B Testing Agile Methodology, AI/ML, Anomaly Detection, Architectural Design, Artificial Intelligence, AWS, CI/CD, Computer Science, Continuous Integration, Database Administrator, Data Gathering, DBA, Deep Learning, DevOps, Developer Tools, Distributed Computing, Docker, ECR, Embedding, ETL, Financial Forecasting, Git, GitHub Actions, GitHub Copilot, Linear Algebra, Large Language Models, LLMs, Mathematics, Multivariate Calculus, Pandas, Power BI, Problem Solving, PyTorch, Redshift, Research, Scikit-learn, S3, Sagemaker, Scripting Language, Scrum, SDLC, Snowflake, SQL, Step Functions Streamlit, Tableau, TensorFlow, Transformers, Version Control

Soft Skills

Adaptability, Business Requirements, Coaching, Creative, Growth Mindset, Leadership, Mentoring, Multicultural Collaboration, Strategic Thinking, Thought Leadership, Willing to Learn